Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	250 N
			Total 1 mark

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Resolves vertically to form an equation to find the tension.	AO3.1b	M1	$0.4 \times 9.8 = T\cos 30°$
	Finds correct tension.	AO1.1b	A1	$T = 4.5$ N
(b)	Finds radius.	AO1.1b	B1	$r = 0.6 \sin 30°$ $= 0.3$
	Forms an equation to find angular speed with 'their' radius and 'their' tension.	AO3.1b	M1	$T = \sin 30° = 0.4 \times r\omega^2$
	Obtains correct angular speed to 2 sf. FT 'their' values provided both M1 marks have been awarded	AO1.1b	A1F	$\omega = \sqrt{\dfrac{4.5 \sin 30°}{0.4 \times 0.3}}$ $= 4.3$ rad s^{-1}
(c)	States two appropriate assumptions.	AO3.5b	B1	Light and inextensible.
				Total 6 marks

Q3.

(a) Resolve vertically:

$T \cos \theta = mg$

M1 for $T \cos\theta$ or $T \sin\theta$ and mg

M1

$34 \cos \theta = 2 \times 9.8$

A1

$\cos\theta = \dfrac{19.6}{34}$

$\theta = 54.8°$

A1

3

(b) Resolve horizontally for particle:

$\dfrac{mv^2}{r} = T\sin\theta$

M1 for $T\cos\theta$ or $T\sin\theta$

M1

$v^2 = \dfrac{34\sin 54.8 \times 0.8}{2}$

A1 ft from (a)

$v^2 = 11.113$

Speed is 3.33 m s^{-1}

Accept 3.34

A1

3

(c) Time taken is $2\pi r / v$

Or find ω and use $\dfrac{2\pi}{\omega}$

M1

= 1.51 sec

A1ft

2

[8]

Q4.

(a) 40 revolutions per minute
= 80π radians per minute

or $\dfrac{2}{3}$ rev per second

B1

$= \dfrac{4\pi}{3}$ radians per second

AG

B1

2

(b) Resolve vertically:

$T \cos 30 = 6g$

M1 1 term each side, 1 correct

M1A1

$T = 67.9$ N

AG

A1

3

(c) Resolve horizontally:

$T \sin 30 = m\omega^2 r$

M1 1 term each side, 1 correct

M1

A1 T sin 30

A1

$67.9 \sin 30 = 6 \times r \times \left(\frac{4\pi}{3}\right)^2$

A1 RHS

A1

$r = 0.322$ m

Condone 0.323 (using π as 3.14)

A1

4

[9]

Q5.

(a) $a = \frac{14^2}{50} = 3.92$

finding acceleration

M1

correct acceleration

A1

$F = 1200 \times 3.92$ AG

use of $F = ma$

dM1

= 4704 N

correct force from correct working

A1

4

(b) R = 1200 × 9.8 = 11760

normal reaction

B1

4704 ≤ μ × 11760

applying $F \le \mu R$

M1

$\mu \ge \frac{4704}{11760}$ AG

correct result from correct working

A1

3

$\mu \le 0.4$

[7]

Q6.

(a) (i) Acceleration = $\frac{v^2}{r}$ = $\frac{(7.5)^2}{15}$

Attempt at $\frac{v^2}{r}$

M1

= 3.75 ms^{-2}

A1

2

(ii) 2940 = 400 × $\frac{V^2}{15}$

M1 use, A1 subs correct

M1A1

V = 10.5 ms^{-1}

A1

3

(b) Motorcycle and rider modelled as a particle

B1

Size of rider/cycle compared with radius / 15 m

B1

2

(c) Acceleration of force $\left(\frac{v^2}{r}\right)$ must decrease

Force decrease → radius increase B1 sc

M1

so r must increase

For 2 marks, algebraic reference or convincing explanation

A1

2

[9]

Q7.

(a) Period = $\frac{2\pi}{\omega}$

M1

$\therefore$ 1.5 = $\frac{2\pi}{\omega}$

$\omega = \frac{2\pi}{1.5} = \frac{4\pi}{3} = 1.3\pi$

Any – must leave π

A1

2

(b) (i) Acceleration = $r\omega^2$

= 0.6 $\left(\frac{4\pi}{3}\right)^3$

Attempt to use formula

M1

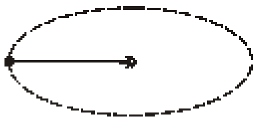
10.5 or $\frac{16\pi^2}{15}$

Follow through their ω

A1ft

2

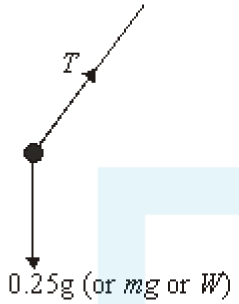
(ii)



B1

1

(c) (i)



(b)(ii) & (c)(i) could be on a single diagram

B1

1

(ii) Vertically (let $A\hat{B}O = \alpha$) $T \sin \alpha = mg$ (1)

Horizontally

Values may or may not be substituted in each equation throughout

M1A1

$T \cos \alpha = mr\omega^2$ (2)

ft $r\omega^2$ from b(i)

M1A1ft

(1) ÷ (2) $\tan \alpha = \frac{g}{r\omega^2}$

Dividing to get tan α or square and add to get T first

M1

$\alpha = \tan^{-1} \dfrac{9.8}{0.6\left(\frac{4\pi}{3}\right)^2}$ or $\tan^{-1}(0.9308)$

$\alpha = 43°$

Rounds to 43°

A1

6

(d) No air resistance
Modelled as a particle

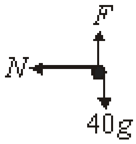
E1

1

[13]

Q8.

(a)



F towards centre scores B0

B1

1

(b) (i) $F = 40g$ or 392 N

B1

1

(ii) $F \leq \mu N$

$392 \leq \mu 784$

use of ≤ or =

M1

$\mu \geq \frac{784}{392} = 0.5$

A1

2

(iii) $N = mr\omega^2$

$mr\omega^2$ seen or used

M1

$784 = 40\,(3)\omega^2$

values substituted

A1

$\omega^2 = 6.5\dot{3}$

$\omega \approx 2.56$

AG

A1

3

(c) Martin modelled as a particle

any suitable assumption

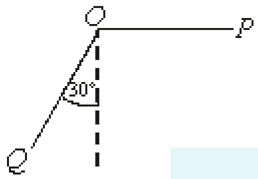
B1

1

[8]

Q9.

(a)



(i) $At\ P$, PE = 70g cos 30° KE = 0

PE or KE term seen correct (Non zero)

B1

$At\ Q$, KE = (70)v^2 PE = 0

Conservation of energy

$\frac{1}{2}(70)v^2$ = 70g5 cos30°

Forming equation

M1A1

v^2 = 10g cos30°

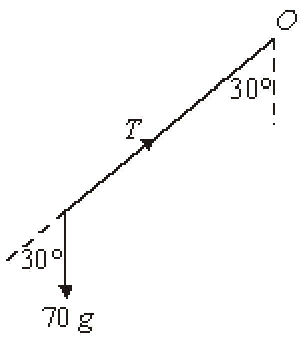
$\Rightarrow$ v = 9.21 ms^{-1}

AG

A1

4

(ii)



Force towards O = T – 70g cos 30°

For circular motion, $F = ma$

$\pm(T - 70g\cos 30°)$ or $\frac{mv^2}{r}$ evaluated

B1

$\Rightarrow T - 70g\cos 30° = \frac{mv^2}{r}$

For equation – Res.force = $\frac{mv^2}{r}$ evaluated

M1A1

5

$\Rightarrow T - 70g\cos 30° = \frac{70(9.21)^2}{5}$

$\Rightarrow T \approx 1782.28....$

$\Rightarrow T = 1780\text{N}$

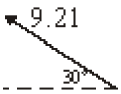
Dependent – substitute and rearrange

m1

AWRT 1780

A1

(b)



$v^2 - u^2 = 2$ as seen

M1

Vertically, $u = 9.21\sin 30°$

Initial vertical component seen = 9.21 sin 30°

A1

$v = 0$

$a = -9.8$

$s = ?$

Using $v^2 - u^2 = 2as$:

$\left(\frac{9.21\sin 30°}{2(9.8)}\right) = 5$

Substitute values into

$v^2 - u^2 = 2as$

M1

$s \approx 1.08$

Approx 1 metre

Must see 1. ...

A1 **4**

(c) Height of a man significant to length if rope/distances involved.
Air resistance would reduce speed / height.

Comment that indicates effect of assumption

B1 **1**

[14]

Q10.

(a) 1100 r.p.m. = 1100 × $\frac{2\pi}{60}$ rad s^{-1}

Allow $\frac{110\pi}{3}$ *OE*

M1

≈ 115 rad s^{-1}

units not required

A1 **2**

(b) 115 = 8$\dot{w}$

Allow $\frac{55\pi}{12}$ *OE*

M1

⇒ $\dot{w}$ ≈ 14.4 rad s^{-2}

units not required

A1 **2**

[4]

Q11.

(a) 2π rads take 27.32 days

2π rads take 27.32 × 24 × 60 × 60 sec

days to seconds

B1

angular speed = $\frac{2\pi}{(27.32 \times 24 \times 60 \times 60)}$

radians per second

M1

≈ 2.66×10^{-6} rad s^{-1}

must see 4sf

A1

3

(b) Using $v = r\omega$:

= $(3.844 \times 10^8) \times (2.66 \times 10^{-6})$

use of formula

M1

≈ 1020 ms^{-1} (3sf)

AWRT

A1

2

[5]

Q12.

(a) $a = 0.6 \times 10^2 = 60$ ms^{-2}

Use of $a = r\omega^2$

M1

Correct acceleration

A1

2

Allow ±60

(b) $R = 0.05 \times 60 = 3$N

Finding product of mass and acceleration

M1

Correct R

A1ft

2

Follow through incorrect a

(c) $R - 0.05 \times 9.8 = 0.05 \times 60$

Equation of motion at lowest point

M1

Correct equation

A1

$R = 3.49$ N

Correct R
Follow through incorrect a

A1ft

3

[7]

Q13.

(a) (i) $R \cos 60° = 3 \times 9.8$

Resolving vertically
Correct equation

M1 A1

$R = 58.8$ N

Correct R

A1

3

(ii) $58.8 \cos 30° = 3 \times \frac{v^2}{0.5}$

Resolving vertically
Correct equation

M1 A1

$v = \sqrt{\frac{58.8 \cos 30°}{6}} = 2.91$ ms^{-1}

Solving for v
Correct v

M1 A1

4

(b) (i) No change

No change

B1

1

(ii) Increased because v^2 is proportional to the radius

Increases
Reason

B1 B1

2

[10]

Q14.

(a) $a = \frac{4^2}{0.1} = 160 \text{ ms}^{-2}$

Calculating the acceleration

M1

Correct acceleration

A1

2

(b) $\cos\theta = \frac{\sqrt{30}}{20} = \frac{\sqrt{3}}{2}$

Identifying trig value

B1

$T_1 \cos\theta = 2 \times 9.8$

Resolving horizontally

M1

Correct equation

A1

$T_1 = \frac{2}{\sqrt{3}} \times 2 \times 9.8 = 22.6 \text{ N}$

Correct force

A1

4

(c) $\sin\theta = \frac{1}{2}$

Identifying trig value

B1

$T_1 \sin\theta + T_2 = 2 \times 160$

Resolving horizontally

M1

Correct equation

A1

$T_2 = 320 - \frac{39.2}{\sqrt{3}} \times \frac{1}{2} = 309 \text{ N}$

Substituting and solving

m1

Correct force

A1

5

[11]

Q3.

In part (a) many students resolved vertically to find the angle but in the required equation $T\cos\theta = mg$ a number of students used $\sin\theta$ rather than $\cos\theta$ whilst others used $mg = \frac{T}{\cos\theta}$.Similar errors were seen in part (b). In part (c) many students found ω and then stopped. The value of ω was not credited since it was just as simple to find the time from knowing only v as it was from knowing ω.

Q4.

In part (a), the conversion of 40 revolutions per minute into $\frac{4\pi}{3}$ radians per second was usually shown correctly. In part (c), candidates often forgot the m in the equation $T\sin 30 = m\omega^2 r$. A common error was in the use of $T\sin 30 = \frac{mv^2}{r}$, where candidates forgot that they were given the value of ω and not the value of r. Candidates were often unable to find correctly the numerical value of r when they had correctly obtained $r = \frac{67.9\sin 30}{6\times\left(\frac{4\pi}{3}\right)^2}$. Some gave the answer as 0.32 and it was surprisingly common for candidates to use $\pi = 3.14$ to find an answer of 0.323. Although this was accepted, it would have been far simpler for candidates to use the π button on their calculators.

Q5.

Part (a) was done very well by the vast majority of the candidates. The printed answer probably helped some of them. In part (b), many candidates were able to obtain the value of 0.4, but far fewer were able to deal correctly with the inequality. When attempting questions like this candidates should start with $F \le \mu R$ and preserve the inequality through out their working. Typically no attempt was made to start with an inequality, which some candidates did then try to introduce at the end of their solution.

Q6.

Part (a) was frequently done well, with the most common error in part (a)(ii) being the introduction into the equation of motion of an additional force, usually the weight of the motorcycle and rider. Part (b) seldom produced assumptions backed by sound reasons. The majority of responses referred to information given in the question. In part (c), the most convincing solutions were based on the algebraic link between the force and the radius. There was a tendency in part (c) to assume that the magnitude of the frictional force could be altered by changing the radius of the circle, rather than the necessity to adjust the radius to compensate for the force.

Q7.

This question provided a better response than was expected. The structure seemed to help candidates to score well. A number of candidates did not seem to be aware of how to find the angular speed given angular displacement and time – part (a) therefore proved discriminating. The correct formula was often seen in part (b), although some substituted incorrectly. Diagrams were excellent and well labelled. Some candidates were well drilled in finding the angle, although some found the angle with the vertical because they had used the angle at the top. Follow through marks meant a critical error early on often only lost

Q8.

There was an excellent response to this question – far better than similar questions in the past - with many candidates scoring 7 or 8 marks. Quite often the force diagram proved problematic with friction incorrectly pointing to the centre of the circle or an additional force existing opposite to the normal reaction. The use of the law of friction was good. Not all candidates were able to give a modelling assumption that was not already stated in the stem. Most common and correct answers here were “treat Martin as a particle” or “air resistance ignored”.

Q9.

Using principles involving energy was well understood and part (a)(i) proved to be very successful. Occasionally some candidates confused themselves by identifying the wrong change in height when calculating potential energy. Part (a)(ii) proved to be far more difficult, many candidates seeming to struggle with this concept. A variety of errors were seen which included: use of $T\cos\theta$ or $T\sin\theta$ in $F = ma$; resultant force in the wrong direction often vertically; and incorrect substitution into mv^2/r. A mixed response was seen to part (b), with a good proportion of candidates realising that the vertical component was required.

Part (c) was more discriminating where candidates who commented on an assumption and considered its likely effect obtained the mark.

Q10.

Most candidates were able to make good progress with this question.

Q11.

This was a good starter. Many candidates clearly knew the relevant formulae well. The accuracy policy meant that many candidates dropped a mark for failing to show a fourth figure in their final calculation of the angular speed.

Q12.

While there were many good solutions to this question, there were also a number of errors that appeared on quite a few scripts.

In part (a) several candidates included the mass when calculating the acceleration.

In part (b) some candidates assumed that the only force was the weight of the sphere.

Finally in part (c), the most common error was to find the difference between the weight and the resultant force, rather than the sum of them.

Q13.

In part (a), almost all of the candidates produced the value of 58.8 N, but quite a few of these did not give a satisfactory argument to support their answer. Some candidates were clearly dividing the weight of the particle by the radius of the circle. In part (b), the common error was to simply use 58.8 without resolving to find the horizontal component. The answers to part (b) were very varied. Some candidates did produce reasonable answers without attempting part (a).

Q14.

The candidates found part (a) straightforward and there were relatively few errors here. A few candidates calculated the resultant force rather than the acceleration. In parts (b) and (c), a few had trouble finding the angles to use when resolving, but most of those who made a reasonable attempt did manage to use the correct angles. There were some errors in forming the equations of motion, often due to either including additional terms or making errors when resolving.